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Assessment of the anxiolytic, antidepressant, and antioxidant potential of *Parquetina nigrescens* (Afzel.) Bullock in Wistar rats

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ABSTRACT

Ethnopharmacological relevance: The recent growing concerns about the multisystemic nature of mental health conditions in the global population are facilitating a new paradigm involving alternative natural, nutritional, and complementary therapies. Herbal remedies despite accounts in literature of their ethnobotanical as alternative remedies for diverse ailments, remain underexplored for psychiatric disorders like anxiety, depression, and insomnia.
Aim of the study: Hence, the anxiolytic, antidepressant, and antioxidant properties of a hydro-ethanolic leaf extract of *Parquetina nigrescens* (PN) in male Wistar rats were investigated.
Materials and methods: The sedative effect was evaluated using the Diazepam sleeping time test while anxiety was induced with a single intraperitoneal injection of 20 mg/kg pentylenetetrazol (PTZ). This was after pre-treatment with 100, 150, and 250 mg/kg of PN or the standard drugs (1 mg/kg diazepam and 30 mg/kg imipramine) for 14 consecutive days. Behavioral tests (Open Field test, Elevated Plus-Maze test, and Forced Swim test) were performed on days 1 and 14, to evaluate the antidepressant and anxiolytic activities of PN. Oxidative stress and neurochemical markers were determined in the brain homogenates of the animals.
Results: The duration of sleep was significantly ($p < 0.001$) increased in the PN-administered group compared to the control. The behavioral models showed that PN exhibited antidepressant and anxiolytic properties in PTZ-induced animals. Significant reductions were observed in GSH level and SOD activity while MDA, nitrite, and GPx levels were significantly increased in PTZ-induced rats. However, treatment with PN significantly improved brain antioxidant status by ameliorating the PTZ-induced oxidative stress. Dopamine, cortisol, and acetylcholine esterase activity levels were significantly ($p < 0.05$) elevated while serotonin and brain-derived neurotrophic factors were reduced in PTZ-induced rats compared with the control.
Conclusion: The PN demonstrated neurotransmitter modulatory ability by ameliorating the PTZ-induced neurochemical dysfunction. Findings from this study showed that PN exhibited sedative, antidepressant, and anxiolytic activities in rats.

1. Introduction

One of the leading causes of occupational redundancy globally is psychiatric disorders like depression and anxiety (Deguchi et al., 2022). According to Charlson et al. (2019), psychiatric disorder is multisystemic with comorbidities that affect one in five people around the world. The World Health Organization (WHO) predicted that by 2030, depression, which is a mood disorder, may become the highest global psychiatric burden in the world (Greenwood et al., 2018). Pessimism is a mood and a sign of depression that affects one's thoughts, feelings, conduct, and sense of well-being (Rosenbaum et al., 2016). Also, suicidal thoughts along with strong feelings of sadness, hopelessness, unhappiness, changes in sleep patterns, lack of appetite, and low energy are clinical signs of depression. Reduced levels of brain neurotransmitters

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like serotonin, dopamine, and norepinephrine are linked to depression in humans. Moreso, patients suffering from depression may produce too much cortisol and experience changes in their circadian rhythm (Fathinezhad et al., 2019). It was estimated that one in three persons may experience depression in economically developed nations while it is logically much higher in countries underdeveloped and experiencing war, political instability, deprivations, civil rights restrictions, and natural disasters. Conversely, anxiety is often expressed in humans as post-traumatic stress, obsessive-compulsive, panic, or social anxiety disorder. It may range from everyday stress to clinically threatening symptoms requiring medical intervention. Anxiety disorder is a serious socioeconomic burden on many societies, and it ranks as one of the most prevalent mental disorders in recent decades (Kabir et al., 2015). The clinical symptoms of anxiety disorder vary and may include heart palpitations, panic feelings, fear, insomnia, sweaty hands and/or feet, shortness of breath, difficulty in remaining calm, a dry mouth, hand or feet numbness or tingling, nausea, tightness in the muscles, and dizziness (Arya and Soodan, 2015). Hence, studies have attributed the pathophysiology of anxiety to the intricate interaction of variables like genetic, developmental, and environmental variables. Biochemical factors like Gamma-aminobutyric acid (GABA) neurotransmission, dopaminergic and serotonin systems, the hypothalamus-pituitary-adrenal (HPA) axis, brain-derived neurotrophic factor (BDNF), and its intracellular signaling pathways are reported by Palazidou (2012) to be pathophysiologically connected to anxiety and depression in humans (Palazidou, 2012) (see Fig. 3).

The comorbidity of anxiety and depression disorders abound in most psychiatric literature and constitutes about 50–60% of psychiatric illnesses worldwide (Hoeflich et al., 2023). Additionally, the overproduction of free radicals or reactive oxygen species (ROS) that characterizes oxidative stress results in the imbalance of pro-oxidants and antioxidants (Afsar et al., 2017). Suffice it to say different degrees of oxidative damage and decline in antioxidant enzymes may occur in people suffering from depression and anxiety.

Selective serotonin reuptake inhibitors (SSRIs), selective serotonin and noradrenaline reuptake inhibitors (SNRIs), tricyclic antidepressants, monoamine oxidase inhibitors (MAOIs), and benzodiazepines (BDZ) *inter alia* may be linked to the current therapies for depression, anxiety, and other related disorders. Although the use of clinical antidepressants has been a common choice for ameliorating moderate, severe, and chronic depression for several decades up until recently when most of them were detected to be less effective (Es-safi et al., 2021). Only 33% of depressed patients are sensitive to the first antidepressant and sedative medications with varying reactions. Some of these side effects may be oral dryness, exhaustion, digestive and respiratory issues, anxiety, agitation, drowsiness, cardiac arrhythmias, psychomotor impairment, decreased alertness, sexual dysfunction, impaired memory, blurred vision, and ataxia (Latha et al., 2015). The range of clinical risks and discomforts linked to clinical antidepressant drugs necessitated the search for alternative treatment options in natural herbs and other plant products for these disorders (Es-safi et al., 2021).

Parquetina nigrescens (Afzel.) Bullock is a shrubby medicinal plant that is commonly referred to as the bullock. It is a member of the family *Ascelpiadaceae* and is found growing luxuriantly in equatorial West Africa. *P. nigrescens* possesses leaves, roots, and latex, which are used in traditional medical practices (Airaodion et al., 2019). These parts have been implicated in treating sexual dysfunction, wounds, anemia, cancer, and occasional insanity for centuries without proven clinical trials (Segun et al., 2018). In Nigeria, *P. nigrescens* was reportedly used as an anti-aging agent and in the management of headaches, insomnia, convulsions, spasms, hypertension, fatigue, pains, and memory loss (Ighodaro et al., 2020). The pharmacological properties of *P. nigrescens* such as antioxidant, anti-ulcer, erythrocyte membrane stabilizer, anti-lipogenic, analgesic, anti-inflammatory, and hematopoietic activities have been reported by Owoyele et al. (2011). However, data on the ethnobotanical and ethnopharmacological applications of this plant as an anxiolytic and antidepressant remain inadequate. Hence, there is a need to investigate its ameliorating effect on anxiety, depression, and stress conditions. This study is therefore designed to screen the ethanolic leaf extract *P. nigrescens* for its sedative, antidepressant, and anxiolytic properties in diazepam (DZP) and pentylenetetrazol (PTZ) induced rats.

2. Materials and methods

2.1. Plant material

Fresh leaves of *P. nigrescens* were collected from Igoba (7.31°N, 5.26°E), Akure, South-Western Nigeria in August 2017. A voucher specimen (FUTA 0242) was placed at the Herbarium after being identified and authenticated by the Department of Crop, Soil, and Pest Management at the Federal University of Technology, Akure (FUTA).

2.2. Experimental animals

Healthy adult male Wistar rats of the Department of Biochemistry, Federal University of Technology, Akure, Nigeria, were fed rodent chow (Vita Feeds, Nigeria limited) and water *ad libitum*, and acclimated under standard conditions for laboratory animal care. The animal protocol adopted for the study was validated by an ethical approval number (FUTA/ETH/21/24) issued by the animal studies subcommittee of the Research Ethics of the Federal University of Technology, Akure.

2.3. Preparation of Parquetina nigrescens hydro-ethanolic leaf extract (PN)

Fresh *P. nigrescens* leaves were properly rinsed free of dirt, dried at room temperature until completely brittle, and then ground into powder using an electric grinder. The powdered leaf (400 g) was macerated in 2 L of 80% v/v ethanol for 24 h and vortexed every 6 h. A muslin cloth was used to filter the suspension, which was then concentrated under decreased pressure in a rotary evaporator before being freeze-dried. After calculating the yield, the remainder was placed in a desiccator for storage.

2.4. Neurobehavioral and biochemical studies

2.4.1. Diazepam sleeping time

In this investigation, the approach outlined by Rakotonirina et al. (2001) was adopted. Twenty adult rats weighing 200 ± 20 g each were randomly selected into four (4) groups. As the control, Group 1 received normal saline; Groups 2, 3, and 4 received 200, 400, and 800 mg/kg of *P. nigrescens* hydro-ethanolic leave extract (PN), respectively. All the rats were given an intraperitoneal dose of 25 mg/kg of diazepam an hour later. The start and length of sleep were then monitored in each rat. The inability of the mice to roll back when flipped over is the requirement for sleep. The interval between loss and recovery of the rightening reflex was used as the index of hypnotic effect.

2.4.2. Evaluation of extracts in PTZ-induced anxiety model

Adult male Wister rats weighing 200 ± 20 g were randomly separated into 11 groups of 6 animals each, and the following treatments were administered to them; The control group, Group 1 got only distilled water. Pentylenetetrazol (PTZ), a common anxiogenic medication, was delivered intraperitoneally (i.p.) for 24 h post-injection to the rats in Group 2 at a dose of 20 mg/kg body weight. *P. nigrescens* hydro-ethanolic leaf extract was given orally to Groups 3, 4, and 5 animals for a total of 14 days at doses of 100, 150, and 250 mg/kg body weight, respectively. Groups 6, 7, and 8 animals were also orally administered 100, 150, and 250 mg/kg body weight of *P. nigrescens* ethanolic leaf extract, respectively, for 14 days using a specialized cannula and then injected PTZ intraperitoneally (i.p) 30 min after administration of the last dosage. The dosages of the ethanolic extract used were based on

LD50 of 5000 mg/kg determined by Lorke's method (Sadiq et al., 2020). Group 9 animals orally received 1 mg/kg Diazepam for 14 days (Uddin et al., 2018). Group 10 animals also received 1 mg/kg Diazepam oral dosage for 14 days and PTZ was intraperitoneally (i.p) administered 30 min after administration of the last dosage. Group 11 rats then received 30 mg/kg of Imipramine orally for 14 days. Rats were subjected to behavioral tests such as Open Field Test (OFT), Elevated Plus Maze (EPM), and Forced Swim Test (FST)) on the first day, and 14th day respectively. This is done 1-h post-treatment on each evaluation day and 30 min after PTZ administration on day 14.

2.5. Behavioral assessment

2.5.1. Open-field test (OFT)

Gellért and Varga (2016) modified approach to Open Field Test (OFT) was carried out. One hour after administration with vehicle (distilled water, 10 ml/kg), extracts (100, 150, and 250 mg/kg), and standard drugs (Diazepam 1 mg/kg and imipramine 30 mg/kg), and 30 min after PTZ (20 mg/kg) administration, each rat was gently placed at the center of the open-field and its behavior monitored and recorded with the aid of a digital chronometer and a video camera mounted above the open-field apparatus for 5 min. This experiment was repeated thrice per treatment group and the number of square crossings which includes when all four paws moved from one square to another (Ambulation), and the number of rearing and assistant rearing (when the fore paws are placed on a wall of the Open field apparatus) were recorded. The floor of the open field was cleaned with ethanol after each session to remove any olfactory cue.

2.5.2. Elevated Plus-Maze test (EPM)

A modified method of Leo et al. (2014), was employed. One hour after administration with vehicle (distilled water, 10 ml/kg), extracts (100, 150, and 250 mg/kg), and standard drugs (Diazepam 1 mg/kg and Imipramine 30 mg/kg), and 30 min after PTZ (20 mg/kg) administration, each rat was placed in turn in the center of the maze facing one of the open arms. The cumulative time spent by each rat in the open and closed arms of the maze, the number of entries into the open and closed arms, and the total entry into both arms were recorded for 5 min, with the aid of a digital chronometer and a video camera mounted above the maze. This process was carried out three times with a different rat from each treatment group and precautions were taken to ensure that the rats were shielded from any external stimulus that could remotely cause anxiety by doing the experiment between 7 a.m. and 10 a.m. A percentage of the total amount of time spent on the open and closed arms was also used to indicate the amount of time on the open arms. Since rodents are prey animals that naturally avoid exposure to the EPM's open arms, a rise in the proportion of time spent there indicated an anxiolytic effect.

2.5.3. Forced swim test (FST)

This test is a valid pharmacological model for assessing antidepressant activity in rodents since its development. Rats were forced to swim in an open cylindrical container from which they could not escape (Yankelevitch-Yahav et al., 2015). The animal behavior was recorded with the aid of a digital chronometer and a video camera mounted above the cylinders. The total amount of time immobile and latency to immobility within 5 min was calculated. The duration of immobility (when the animal ceases struggling and remains floating), when exposed to an inescapable situation (FST), reflects depression. According to this depression model, depression gets worse the longer the rats remain immobile. Hence, the behavioral responses to antidepressant activities were measured by decreased immobility and an increase in struggle time.

2.6. Biochemical evaluation

The rats were sacrificed by cervical dislocation 24 h after the last treatments. Blood samples were collected by cardiac puncture into plain bottles and allowed to clot. The serum was aspirated from the supernatant after the blood was centrifuged at 3000 rpm for 10 min at 4 °C after which it was stored at -4 °C for biochemical studies. The brain was also excised, rinsed in an icy potassium chloride solution of 1.15 percent (w/v), blotted on filter paper, and weighed (Paasonen et al., 2022). The organ was then homogenized in 10% (w/v) phosphate buffer saline (pH 7.4) to get the supernatant, which was then stored at -4 °C for biochemical analyses, the resultant homogenate was centrifuged at 10 000×*g* at 25 min for 4 °C.

2.7. Oxidative stress and antioxidant markers

The level of reduced glutathione (GSH) was determined using the technique of Beutler et al. (1985). Superoxide dismutase (SOD) activity in the homogenates was assessed using the Kakkar et al. (1984) technique. Glutathione peroxidase activity was determined according to the method of Haque et al. (2003). The approach of Varshney and Kale (1990) was used to measure the production of thiobarbituric acid reactive substances (TBARS) to determine the level of lipid peroxidation. Similarly, nitrite and nitrate evaluation in biological materials is used as the marker for nitric oxide (NO) generation. The nitrite levels were estimated in the brain homogenates using the Griess reaction (Guevara et al., 1998).

2.8. Assessment of neurotransmitter dysregulation

The level of dopamine was evaluated using the method of Guo et al. (2009). With a few minor adjustments, the spectrophotometric approach of Ellman et al. (1961) was used to quantify the activity of acetylcholinesterase (AChE), using the substrate acetylcholine iodide. Serotonin level was estimated using the Serotonin ELISA (Enzyme-Linked Immunosorbent Assay) kit 96 T (Elabscience Biotechnology Co. Ltd.). The results are expressed as ng of serotonin/ml of tissue. Cortisol level was estimated using Cortisol ELISA (Enzyme-Linked Immunosorbent Assay) kit 96 T (Calbiotech, Inc [CBI]). The results are expressed as ng of cortisol/ml of tissue. Using the rat BDNF ELISA (Enzyme-Linked Immunosorbent Assay) kit 96 T (Elabscience Biotechnology Co. Ltd.), the level of Brain-derived neurotrophic factor (BDNF) was estimated. The results are expressed as pg of BDNF/ml of tissue.

2.9. Statistical analysis

The mean and standard deviation (SD) of the results were reported. Microsoft Excel and GraphPad Prism 8.0 (GraphPad Software Inc., CA, USA) were used to analyze the experimental data. One-way analysis of variance (ANOVA) and Tukey's multiple comparison tests were used for the statistical studies. $p < 0.05$ was used as the criterion for statistical significance in each test.

3. Results

3.1. Effect of PN on neurobehavioral indices in rats

3.1.1. Effect of extract on diazepam sleeping time of rats

The extract at 400 and 800 mg/kg increased statistically ($p < 0.0001$) the duration of sleep of the animals but had no significant effect on the latency to sleep time (Fig. 1). The sleep duration of the control group (19.59 ± 1.99 min) was surpassed by a factor of 2 and 3 in the treated groups, PN 400 and 800 (38.0 ± 5.00, 58.5 ± 3.50), respectively (see Fig. 2).

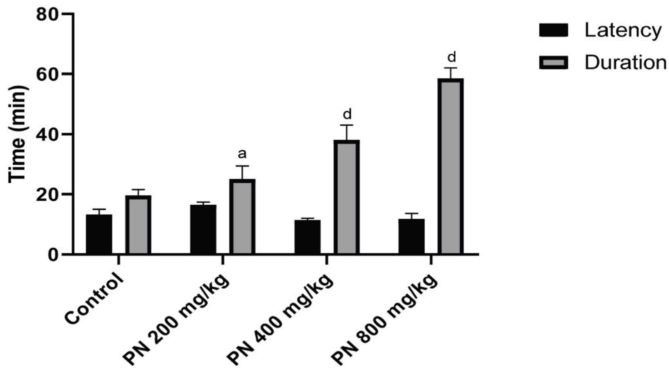

Fig. 1. Effect of *P. nigrescens* ethanolic (PN) leaf extract on Diazepam sleeping time. Results are expressed as Mean ± SD (n = 5). Bars with different alphabets differ by Tukey's test at $^{a}p < 0.05$; $^{d}p < 0.0001$ vs. Control.

3.1.2. *Effect of extract in open field test*

The extract demonstrated anxiolytic activity in the open field test (Table 1). On acute administration (Day 1), the extract showed a significant ($p < 0.05$, 0.0001) increase in the ambulation of the animals compared with the control group (36.60 ± 3.06), with a peak effect noticed at 100 mg/kg dose (62.67 ± 4.95) which was comparable with that of the Diazepam treated group (59.00 ± 10.39). The central crossing also increased (9.00 ± 1.41) at this dose compared with the control (3.50 ± 1.00), and the increase made by diazepam. Meanwhile, PN 250 mg/kg demonstrated a significant ($p < 0.01$) reduction in rearing compared with the control (6.00 ± 1.41 and 11.50 ± 2.38, respectively). Chronic administration of the ethanolic extract of *P. nigrescens* leaf for 14 days also revealed anxiolytic effects. There was a significant ($p < 0.001$, $p < 0.0001$) increase in ambulation and central crossing by the PN at 150 and 250 mg/kg, respectively. Diazepam showed no increase in the ambulation of the rats after 14 days, but a significant ($p < 0.0001$) increase was observed in central crossing (4.33 ± 0.71) compared with the control (1.00 ± 0.10). PN 250 mg/kg significantly ($p < 0.001$) increased rearing while PTZ increased the animal's locomotory ability compared with the control. PN increased significantly ($p < 0.01$, 0.001, 0.0001) the ambulation in the PTZ-induced rats and central crossing at 100 mg/kg. PN (150 and 250 mg/kg) also reduced rearing in the PTZ-induced rats.

3.1.3. *Effect of extract in elevated plus maze test*

The results suggest that the extract has anxiolytic properties. The extract and the drugs (diazepam and imipramine) showed statistically significant ($p < 0.0001$, 0.001) increase in the total entry into the arms and open-arm entry of the rats from the first day compared with the control, while a reduced closed arm entry was equally observed (Table 2). The open arm time was increased by the extract at 100 mg/kg (23.22 ± 5.53) when compared with the control (11.42 ± 1.45). The same trend was observed on day 14, with the extract, diazepam, and imipramine exhibiting anxiolytic activity as supported by observed increase in the total entry into the arms, open arm entry, and open arm time relative to the control group. There was also a corresponding

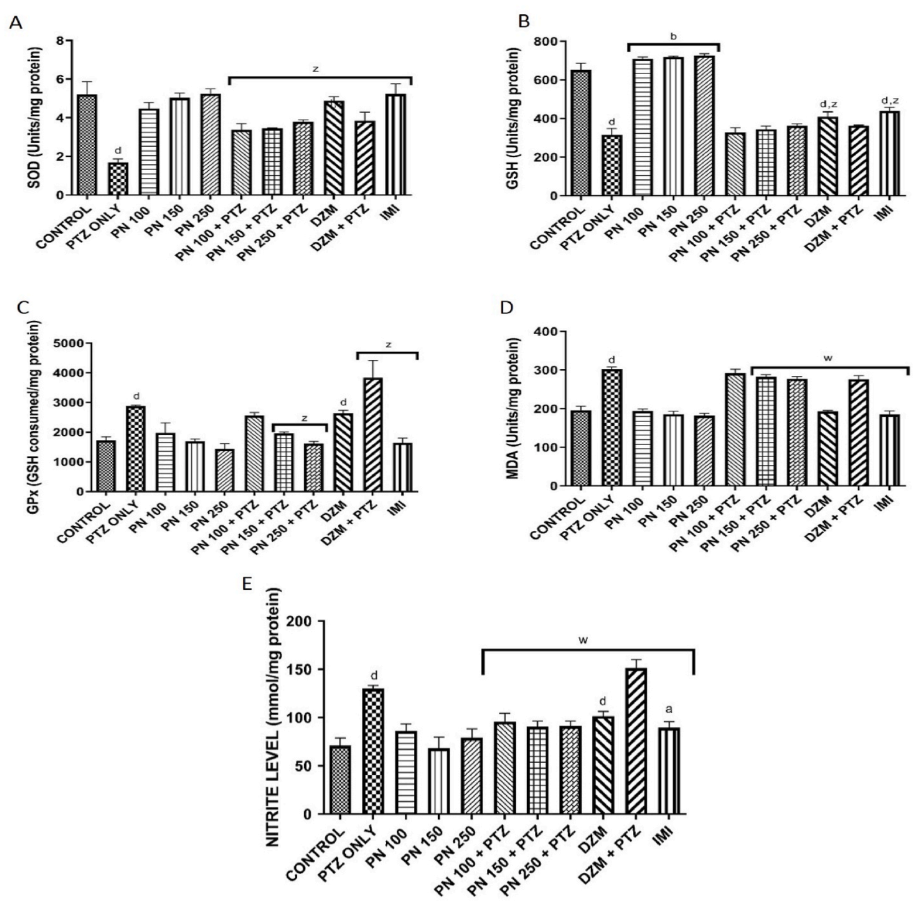

Fig. 2. Effect of PN on oxidative stress biomarker dynamics in the cerebral homogenates of rats. A-SOD, B-GSH, C-GPx, D-MDA, E-nitrite level. Results are expressed as Mean ± SD (n = 6). $^{d}p < 0.0001$, $^{b}p < 0.01$, $^{a}p < 0.05$ vs. CONTROL. $^{z}p < 0.0001$, $^{w}p < 0.05$ vs. PTZ ONLY. Abbreviations: CONTROL: normal saline; PTZ ONLY: pentylenetetrazole (20 mg/kg); PN 100, 150, 250: *P. nigrescens* leaf extract at doses of 100, 150 and 250 mg/kg respectively; PN 100 mg/kg, 150 mg/kg, 250 mg/kg + PTZ: PN at doses of 100, 150 and 250 mg/kg respectively followed by PTZ on day 14; DZP ONLY: Diazepam (1 mg/kg); DZP + PTZ: Diazepam (1 mg/kg) followed by PTZ on day 14; IMI: Imipramine (30 mg/kg).

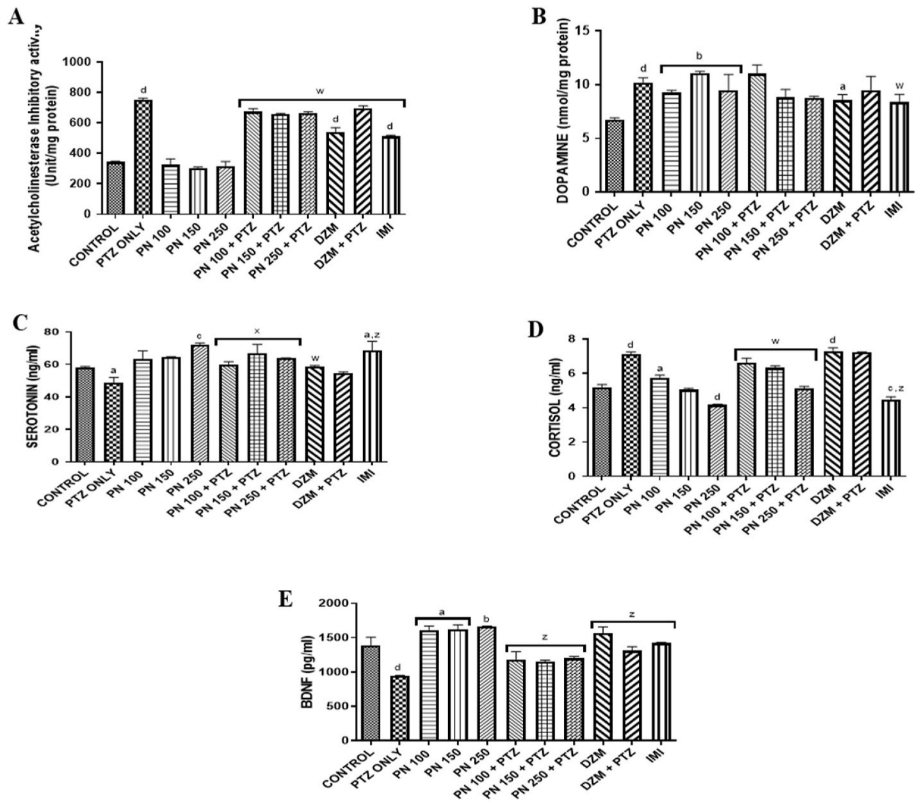

Fig. 3. Effect of *P. nigrescens* ethanolic leaf extract on neurophysiologic dynamics in the brain homogenates of rats. A-Acetylcholinesterase (AChE) Inhibitory Activity, B-Dopamine (DA), C-Serotonin, D—Cortisol, E-BDNF. Results are expressed as Mean ± SD (n = 6). $^{d}p < 0.0001$, $^{c}p < 0.001$, $^{b}p < 0.01$, $^{a}p < 0.05$ vs. CONTROL. $^{z}p < 0.0001$, $^{w}p < 0.05$ vs. PTZ ONLY. Abbreviations: CONTROL: normal saline; PTZ ONLY: pentylenetetrazole (20 mg/kg); PN 100 mg/kg, 150 mg/kg, 250 mg/kg: PN at doses of 100, 150 and 250 mg/kg respectively; PN 100 mg/kg, 150 mg/kg, 250 mg/kg + PTZ: PN at doses of 100, 150 and 250 mg/kg respectively followed by PTZ on day 14; DZP ONLY: Diazepam (1 mg/kg); DZP + PTZ: Diazepam (1 mg/kg) followed by PTZ on day 14; IMI ONLY: Imipramine (30 mg/kg).

decrease in closed arm entry and closed arm time in the groups (Table 2b).

PTZ anxiogenic activity was reflected by decreased values of open arm entry (40.69 ± 3.10 and 48.00 ± 3.47, respectively), and open arm time (2.83 ± 0.69 and 4.61 ± 0.64, respectively) compared with the control. Also, a statistically significant ($p < 0.0001$) increase was observed in the closed arm entry (59.31 ± 3.10 and 51.89 ± 3.47, respectively), and closed arm time (97.17 ± 1.01 and 95.39 ± 1.45, respectively). The extract, diazepam, and imipramine were able to increase the total entry when compared with the PTZ group. Also, PN 250, diazepam, and imipramine showed statistically significant ($p < 0.01$) increase in open arm entry when compared with the PTZ group. Likewise, the anxiolytic effect of the extract was further affirmed by the increase in the open arm time for all the doses in the PTZ-induced rats, which compared with that of diazepam.

3.1.4. Effect of extract in forced swim test

The effects of the PN, PTZ, diazepam, and imipramine on FST were observed to reduce the immobility time and increase the latency to immobility in comparison to the control group (Table 3). The chronic administration of the extract and drugs also followed the same trend. PTZ demonstrated a depressogenic effect with a statistically significant ($p < 0.0001$) increase in immobility when compared with the control (83.25 ± 7.97 and 67.50 ± 3.30, respectively), and a nominal decrease in latency to immobility, compared with the control (30.25 ± 4.62 and 39.00 ± 2.12, respectively).

3.2. Effect of PN on oxidative stress parameters

PTZ caused a statistically significant ($p < 0.0001$) decrease in the GSH, and SOD activity compared with the control, while a significant ($p < 0.0001$) increase was recorded in the GPx, nitrite level, and MDA activity.

3.3. Effect of the PN on neurological parameters

PTZ elicited significant ($p < 0.0001$) increases in dopamine level, acetylcholinesterase activity, and cortisol level, with a significant ($p < 0.05$, $p < 0.0001$, respectively) decrease in serotonin and BDNF levels when compared with the control. The treatment group (PN + PTZ) demonstrated an ameliorative effect on acetylcholinesterase activity. It also increased serotonin levels, reduced cortisol levels, and increased BDNF levels in the treatment groups. Like diazepam, the extract demonstrated a significant ($p < 0.01$) increase in dopamine levels when compared to the control, while acetylcholinesterase activity did show any significant difference. The extract (250 mg/kg) and imipramine significantly ($p < 0.001$, 0.05, respectively) increased the serotonin level compared with the control, while the cortisol levels were significantly decreased. The extract also increased BDNF levels compared with the control group.

4. Discussion

Animal models are often explored in investigating the neurobiological mechanisms underpinning sleep and mood disorders, as well as in the discovery of new therapeutic interventions of natural origins. In West Africa, *P. nigrescens* was reported in the traditional management of neuropsychiatric conditions like insanity, insomnia, depression (Iniaghe et al., 2017), and cognitive dysfunction (Ademosun et al., 2022). Empirical studies in support of these ethnobotanical reports are

Table 1
Effect of *P. nigrescens* ethanolic leaf extract on behavioral parameters in the open field test.

GROUP	Ambulation		Center crossing		Rearing	
	Day 1	Day 14	Day 1	Day 14	Day 1	Day 14
CONTROL	36.60 ± 3.06	17.67 ± 5.51	3.50 ± 1.00	1.00 ± 0.10	11.50 ± 2.38	2.60 ± 0.50
PTZ ONLY	34.67 ± 2.89	30.00 ± 1.41^{d}	4.50 ± 1.00	2.50 ± 0.71^{b}	12.67 ± 2.07	4.40 ± 0.58^{b}
PN 100	62.67 ± 4.95^{d}	13.67 ± 2.31	9.00 ± 1.41^{d}	1.40 ± 0.55	13.67 ± 1.15	3.33 ± 0.31
PN 150	42.33 ± 2.52	27.00 ± 2.36^{b}	4.00 ± 1.00	2.75 ± 0.55^{c}	8.75 ± 1.50	3.75 ± 0.58
PN 250	48.50 ± 6.36^{a}	28.00 ± 3.54^{c}	3.00 ± 0.50	4.33 ± 0.71^{d}	6.00 ± 1.41^{c}	5.00 ± 0.71^{d}
PN 100 + PTZ	59.67 ± 3.21^{d}	54.80 ± 5.48^{z}	11.00 ± 0.82^{d}	4.67 ± 0.71^{z}	10.50 ± 2.12	3.33 ± 1.00
PN 150 + PTZ	40.33 ± 4.69	38.60 ± 1.73^{x}	3.00 ± 0.82	3.00 ± 0.82	8.50 ± 1.71	2.50 ± 0.58^{y}
PN 250 + PTZ	46.60 ± 5.77	41.00 ± 1.73^{y}	3.00 ± 1.00	3.00 ± 0.45	9.83 ± 2.38	2.25 ± 0.58^{z}
DZP ONLY	59.00 ± 10.39^{d}	24.25 ± 2.12	13.00 ± 1.41^{d}	4.33 ± 0.71^{d,y}	14.33 ± 2.52	4.00 ± 0.58^{a}
DZP + PTZ	53.00 ± 7.82^{c}	38.20 ± 2.51^{x}	14.17 ± 0.84^{d}	3.50 ± 0.84	14.17 ± 2.86	3.00 ± 0.71^{w}
IMI ONLY	39.60 ± 4.45	26.67 ± 6.26^{b}	10.33 ± 1.29^{d}	3.25 ± 0.58^{d}	10.33 ± 1.29	4.67 ± 1.09^{c}

Results are expressed as Mean ± SD (n = 6). $^{d}p < 0.0001$, $^{c}p < 0.001$, $^{b}p < 0.01$, $^{a}p < 0.05$ vs. CONTROL. $^{z}p < 0.0001$, $^{y}p < 0.001$, $^{x}p < 0.01$, $^{w}p < 0.05$ vs. PTZ ONLY. Abbreviations: CONTROL: normal saline; PTZ ONLY: pentylenetetrazole (20 mg/kg); PN 100 mg/kg, 150 mg/kg, 250 mg/kg: PN at doses of 100, 150, and 250 mg/kg respectively; PN 100 mg/kg, 150 mg/kg, 250 mg/kg + PTZ: PN at doses of 100, 150 and 250 mg/kg respectively followed by PTZ on day 14; DZP ONLY: Diazepam (1 mg/kg); DZP + PTZ: Diazepam (1 mg/kg) followed by PTZ on day 14; IMI ONLY; Imipramine (30 mg/kg).

Table 2
Effect of *P. nigrescens* Leave hydro-Ethanolic Extract on Behavioral Parameters.

GROUP	TOTAL ENTRY		OPEN ARM ENTRY (%)		CLOSED ARM ENTRY (%)	
	DAY 1	DAY 14	DAY 1	DAY 14	DAY 1	DAY 14
CONTROL	3.67 ± 0.55	2.67 ± 0.45	42.78 ± 7.26	48.00 ± 3.47	57.22 ± 7.26	51.89 ± 3.47
PTZ ONLY	3.67 ± 0.58	5.67 ± 0.71^{d}	38.89 ± 5.77	40.69 ± 3.10	61.11 ± 5.77	59.31 ± 3.10
PN 100	8.50 ± 1.15^{d}	4.33 ± 0.50^{c}	59.73 ± 5.34^{d}	63.33 ± 6.12^{c}	40.27 ± 5.34^{d}	39.44 ± 3.65^{a}
PN 150	7.00 ± 0.58^{d}	3.67 ± 0.58	50.74 ± 2.48	51.39 ± 9.13	49.26 ± 2.48	48.61 ± 9.13
PN 250	6.50 ± 0.45^{d}	4.17 ± 0.45^{b}	64.46 ± 4.45^{d}	63.24 ± 4.d1^{c}	35.54 ± 6.39^{d}	36.76 ± 6.42^{b}
PN 100 + PTZ	7.67 ± 1.15^{d}	8.67 ± 0.51^{z}	57.98 ± 6.63^{c}	49.44 ± 5.24	42.02 ± 5.15^{c}	51.56 ± 5.24
PN 150 + PTZ	7.00 ± 0.58^{d}	6.67 ± 1.00	53.28 ± 4.91^{a}	39.50 ± 6.35	46.71 ± 4.91^{a}	60.50 ± 6.35
PN 250 + PTZ	7.67 ± 0.58^{d}	7.50 ± 0.50^{z}	65.96 ± 6.44^{d}	56.43 ± 3.75^{y}	34.04 ± 6.44^{d}	43.57 ± 6.40^{y}
DZP ONLY	11.67 ± 1.15^{d}	6.33 ± 0.50^{d}	56.09 ± 5.78^{b}	60.08 ± 2.16^{b,z}	43.92 ± 5.78^{b}	39.92 ± 2.16^{a,z}
DZP + PTZ	10.83 ± 1.00^{d}	8.33 ± 0.58^{d,z}	58.09 ± 5.15^{c}	52.86 ± 4.12^{x}	41.91 ± 5.15^{c}	47.14 ± 5.08^{w}
IMI ONLY	5.67 ± 0.71^{b}	7.33 ± 0.55^{d,y}	40.28 ± 2.94	56.84 ± 5.70^{z}	59.72 ± 2.95	43.15 ± 8.78^{y}

Results are expressed as Mean ± SD (n = 6). $^{d}p < 0.0001$, $^{c}p < 0.001$, $^{b}p < 0.01$, $^{a}p < 0.05$ vs. CONTROL. $^{z}p < 0.0001$, $^{y}p < 0.001$, $^{x}p < 0.01$, $^{w}p < 0.05$ vs. PTZ ONLY. Abbreviations: CONTROL: normal saline; PTZ ONLY: pentylenetetrazole (20 mg/kg); PN 100 mg/kg, 150 mg/kg, 250 mg/kg: PN at doses of 100, 150 and 250 mg/kg respectively; PN 100 mg/kg, 150 mg/kg, 250 + PTZ: PN at doses of 100, 150 and 250 mg/kg respectively followed by PTZ on day 14; DZP ONLY: Diazepam (1 mg/kg); DZP + PTZ: Diazepam (1 mg/kg) followed by PTZ on day 14; IMI ONLY; Imipramine (30 mg/kg).

Table 3
Effect of *P. nigrescens* ethanolic leaf extract on behavioral parameters in the forced swim test.

GROUP	TIME IMMOBILE (%)		LATENCY TO IMMOBILITY (S)	
	Day 1	Day 14	Day 1	Day 14
CONTROL	73.00 ± 5.19	67.50 ± 3.30	45.00 ± 7.57	39.00 ± 2.12
PTZ ONLY	74.40 ± 3.02	83.25 ± 7.97^{c}	44.80 ± 1.18	30.25 ± 4.62
PN 100	55.20 ± 4.07^{d}	42.00 ± 3.75^{d}	68.33 ± 3.51^{d}	85.00 ± 4.24^{d}
PN 150	51.67 ± 1.65^{d}	53.33 ± 2.33^{b}	52.75 ± 5.51	53.50 ± 5.86^{c}
PN 250	58.55 ± 3.10^{d}	46.17 ± 0.71^{d}	55.00 ± 6.24	58.00 ± 1.06^{d}
PN 100 + PTZ	56.22 ± 2.69^{d}	66.22 ± 2.33^{y}	60.60 ± 5.50^{b}	42.00 ± 3.77^{w}
PN 150 + PTZ	44.83 ± 4.95^{d}	66.67 ± 1.00^{y}	53.25 ± 9.80	40.00 ± 5.89
PN 250 + PTZ	59.67 ± 5.42^{d}	65.40 ± 10.45^{z}	55.00 ± 2.83	30.00 ± 3.56
DZP ONLY	40.67 ± 4.44^{d}	56.28 ± 4.01^{a,z}	104.83 ± 8.96^{d}	90.33 ± 2.12^{d,z}
DZP + PTZ	40.22 ± 3.03^{d}	66.50 ± 7.31^{y}	102.67 ± 7.50^{d}	61.25 ± 7.78^{z}
IMI ONLY	60.78 ± 3.18^{d}	34.33 ± 9.14^{d,z}	60.50 ± 5.03^{b}	132.75 ± 10.61^{d,z}

Results are expressed as Mean ± SD (n = 6). $^{d}p < 0.0001$, $^{c}p < 0.001$, $^{b}p < 0.01$, $^{a}p < 0.05$ vs. CONTROL. $^{z}p < 0.0001$, $^{y}p < 0.001$, $^{w}p < 0.05$ vs. PTZ ONLY. Abbreviations: CONTROL: normal saline; PTZ ONLY: pentylenetetrazole (20 mg/kg); PN 100, 150, 250: *P. nigrescens* leaf extract at doses of 100, 150 and 250 mg/kg respectively; PN 100, 150 250 + PTZ: *P. nigrescens* leaf extract at doses of 100, 150 and 250 mg/kg respectively followed by PTZ on day 14; DZP ONLY: Diazepam (1 mg/kg); DZP + PTZ: Diazepam (1 mg/kg) followed by PTZ on day 14; IMI ONLY; Imipramine (30 mg/kg).

inadequate and require more scientific research.

Sleep disorder is rising among different demographic populations, particularly among the global adult population, and becoming a public health concern. Adequate sleep is critical to a broad range of physiological functions associated with metabolic and endocrine health, brain activity, emotional stability, and immune system function (Chung et al., 2023). Even though many studies linked disruptive sleep conditions to stress factors, the use of natural herbs, ayurveda, complementary (yoga, acupuncture, massage, and aerobic exercises), and nutrition therapies are gaining acceptance in their management (Ong and Manber, 2023). In Africa, herbs, natural fruits, local beverages, and foods are often used to treat people with insomnia and other health issues. This knowledge heritage became more acceptable because the therapeutic agents are accessible, affordable, and presumed safe without any side effects, and dosage concerns (Inoue et al., 2022). In this study, the dose-dependent PN demonstrated a sedative effect and prolonged the duration of sleep diazepam-induced rats in the sleep time test. Like *Valeriana* sp. (valerian), *Matricaria* sp. (chamomile), and *Humulus lupulus* (hops), the PN sedative effect could be linked to the interplay of biochemical processes involving modulation of adenosine receptors, melatonergic system, and GABAergic activity (Moon et al., 2022). The sedative-hypnotic agents inherent in these plants either directly activate the Gamma Amino Butyric Acid (GABA) receptors or more frequently, enhance GABA's effect on Gamma Amino Butyric A-receptors (GABAA) to mediate synaptic inhibition. This suggests that the PN effect may be compared to some common sleep prescription medications (benzodiazepines and barbiturates) which are positive allosteric modulators of GABA A-receptors (Olsen, 2018). Suffice it to say the PN interferes with GABA-mediated synaptic transmission to enhance the sedative effects of diazepam in the diazepam-induced rats. While the specific constituent or nature of synergistic interaction underpinning the sedative effect is not yet understood, Bruder et al. (2020) linked it to the phytochemical. Hence, further studies on the sedative, anxiolytic, and antinociceptive properties of the PN might potentially improve its commercial as well as commodification values (Adase et al., 2022).

The qualitative advantage of the open field test (OFT) in assessing emotional disorders like anxiety, locomotor activity, sedative, toxic, and stimulant effects of plant extracts or compounds has made it attractive in behavioral studies. The tendency of animals to naturally express behavioral changes when removed from a familiar environment to a strange one gave credence to this test. However, such animals tend to be less mobile and less inclined to explore, more protective of their young, and more disposed to grooming. There could also be signs of increased micturition and defecation due to increased autonomic activity or fear. This study showed there was an increased number of square crossings (ambulation), center crossing, and rearing/assistant rearing compared with the control group which suggests that the PN is anxiolytic. The PN at a concentration of 100 mg/kg was more anxiolytic than at higher doses in the OFT on day 1, while on day 14, the modulation was observed to be dose-dependent. While the reason for the observed trend is unknown, the environment, sensitivity, and paradoxical (false positive) reactions may have accounted for it (Jaiswal et al.,1994). Pentylenetetrazol (PTZ) is anxiogenic in humans, and its interoceptive effects are also expressed in rats. In matured rats treated with PTZ, the GABAA receptor inhibitor depolarizes the animals' neurons inhibiting GABA-mediated action and decreasing the multiunitary activity of hippocampal neurons (Romero-Guerrero et al., 2022). Therefore, PTZ neuronal excitability as observed in the OFT results may be due to increased activation of the glutamatergic/stimulatory system and GABA inhibition. Increased ambulation and center crossing with a reduction in rearing were observed in PTZ-induced rats treated with PN. However, more investigations are required to understand if PN modulated the inhibitors or agonists of PTZ in the induced rat group to cause the observed motor functions, which demonstrate cognitive reflexes rather than anxiolysis (Uddin et al., 2018).

Animals subjected to the elevated plus maze (EPM) test exhibited stronger approach-avoidance conflict than when exposed to an enclosed arm model. Moreso, an increase in the time spent at entry into open arms and a decrease in the aversion to open arms are signs of anxiolysis. The significantly increased total entry, open arm entry, and open arm time by PN-treated rats in the EPM test suggest an anxiolytic property. Anxiogenic drugs, like PTZ, normally reduce animals' open arm entry and open arm time in the EPM test, while conversely increasing the closed arm entry and closed arm time. Thus, PTZ (20 mg/kg) induced anxiety in rats used for this study. The extract, just like the standard anxiolytic drug Diazepam, reduced the closed arm entry and closed arm time while increasing the open arm entry and open arm time in the induced rats. Conversely, depression is another behavioral change or psychiatric condition characterized by the human inability to manage or cope with stress (Armario, 2021). Many antidepressants (ADs) decrease immobility in the FST when taken as a pharmacological treatment before the test. Therefore, this model could assess coping strategies in rats exposed to inescapable stress situations. In this test, immobility is an indicator of despair in the rats. A lack of optimism, and low mood, are also indicators of depressive behaviors (Afsar et al., 2017). PTZ was observed to prolong immobility periods and shorten immobility latency in experimental animals, which suggests that it may be depressogenic to rats. The FST results showed that the PN ameliorated PTZ-induced depressive conditions in the animals (Armario, 2021). Even though more studies are required to sufficiently understand the biochemical mechanisms underscoring this bioactivity, Luo et al. (2021) have linked it to interference caused by some bioactive agents to the rats' neurotransmitters, neuroendocrine, and hippocampal responses. Traditional medicinal plants including *P. nigrescens*, are known to harbor a wide variety of bioactive agents like alkaloids, flavonoids, and tannins that may interactively exert mood-modulatory, sedative, and anxiolytic effects (Phootha et al., 2022). It was observed that the antidepressant effect of some medicinal plants may be potentiated by the synergetic and reactive nature of their bioactive constituents. This mediates, directly or indirectly, monoamine oxidase inhibition, and serotonin (5-HT), dopamine (DA), and noradrenaline (norepinephrine) levels in certain brain areas. Hence, the antidepressant activity of the PN may be attributed to its positive modulation of serotoninergic and dopaminergic neurotransmission in the brain of the experimental rats.

Oxidative stress occurs if ROS generation outpaces a cell's ability to scavenge free radicals while a balance of this function protects physiological tissue damage. Superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx) are significant antioxidant enzymes that play crucial roles in modulating and converting free radicals (Rodriguez et al., 2013). Similar functions are attributed to non-enzymatic glutathione and uric acid antioxidants. Khedr et al. (2015) observed that PTZ increases lipid peroxidation (malondialdehyde-MDA) in the serum and liver of rats causing oxidative stress-induced damage while decreasing SOD and non-enzymatic antioxidant activity. This observation concurred with reduced SOD and glutathione (GSH), and the increased MDA observed in this study after PTZ administration, suggesting a decrease in antioxidant enzymes in the depressive rats. The stimulation of membrane phospholipases, proteases, and nucleases are just a few of the biochemical reactions linked to PTZ activity, and their role in increased oxidative stress as demonstrated by this study. Furthermore, the elevation in the levels of protein oxidation, lipid peroxidation, and inflammatory mediators in the hippocampus and cerebral cortex of PTZ-administered rats have been reported by Khedr et al. (2015). The results from this study showed that the biochemical alterations in PTZ-administered rats were ameliorated by the PN as indicated by reduced values of MDA, nitrite, and GPx, and increased levels of GSH and SOD in the induced rats. The potential of the extract at attenuating oxidative stress in PTZ-induced rats, and the increased levels of SOD and GSH observed confirmed that the plant has potent antioxidant and neuroprotective properties. Relative to the control, the extract increased significantly ($p < 0.01$) the GSH level, without any significant alteration in the SOD, GPx, and nitrite levels. Oxidative stress in the PTZ-induced rats was mediated by the PN, which hypothetically lowers the risk of tissue damage.

The most versatile member of the neurotrophin family, brain-derived neurotrophic factor (BDNF), promotes the migration and differentiation of glial, peripheral, and central neurons in mature rats. Also, it sustains adult growth, survival, and synaptic plasticity, which influences learning and memory. Czeh et al. (2016) affirmed the use of animal models in assessing the cause-and-effect relationship between BDNF, therapies, depressive symptoms, and pathology in stress-related (foot shock, forced swims, immobilization, maternal deprivation, restraint, social isolation, social defeat, and chronic stress studies. BDNF decreased levels indicate a vulnerability to stress and depression by the rats. Reports of some preclinical studies have revealed that reduced BDNF synthesis and protooncogene TrkB activity in the hippocampus and frontal cortex are linked to chronic stress, depressive-like symptoms, and anxiety disorders (Giacobbo et al., 2019). Therefore, one of the depressogenic evidence in the animals subjected to the forced swim test is a decreased BDNF mRNA level in the hippocampus, which affirms the prospect of BDNF in antidepressant therapy. Many antidepressant medications have been suggested to have BDNF signaling as a downstream target (Bjorkholm and Monteggia, 2016). Hence, matured rats that received direct BDNF injections into their hippocampus or midbrain may experience antidepressants, increased neurogenesis, and regional neuronal activities. Also, chronic corticosterone administration lowered BDNF levels in another study by Paizanis et al. (2010). Interestingly, treatment with the PN was observed to reverse the PTZ effect and suggests that the PN could biochemically activate the BDNF signaling cascade to bolster oxidative stress response, apoptosis, and inflammation in the experimental rats (Chen et al., 2023). This is in addition to its ability to potentially upregulate hippocampal neurogenesis and cell survival rates and regulate synaptic/neural plasticity. The temporal effects of BDNF are apparent measures of the temporal profile of clinical antidepressants (Caffino et al., 2024).

The hypothalamic-pituitary-adrenal (HPA) axis dysfunction is also linked to mood disorders such as anxiety and depression. This normally results in increased secretion of corticosteroids (Cortisol and corticosterone) from the adrenal cortex. As a systemic intercellular signal, the corticosterone level rises dynamically in response to stressful situations (Spencer and Deak, 2017). In biological systems, glucocorticoids interact with receptors in the hippocampus and hypothalamus and play a self-modulatory role in their feedback inhibition. In humans, for instance, decreased motor responsiveness and anxiety behavior have been associated with elevated cortisol levels (Patidar and Shaikh, 2013). Comparatively, rats' blood cortisol levels were observed to rise dramatically after PTZ treatment. Reduction in BDNF level and increase in cortisol level are strong indicators of PTZ-induced neurological dysfunction, and these phenomena were mitigated by the PN treatments. Budziszewska et al. (2000) noted that antidepressants block glucocorticoid receptor (GR)-mediated gene transcription which may underscore the anxiolytic and antidepressive effects of the PN. This is because antidepressants raise GR levels and improve GR-mediated feedback inhibition, leading to lower cortisol levels and prevention of cortisol from gene transcription activity (Anacker et al., 2011).

Imbalances in monoamines (NA, DA, and 5-HT) are linked to momentary behavioral anomalies involving memory, learning, moods, appetite, and sleep. Stressful circumstances are some of the main causes of memory loss which is due to the deregulated function of monoamines (Patidar and Shaikh, 2013). Antidepressants increase BDNF production, and BDNF improves monoamine neurons' ability to survive by raising the levels of neurotransmitters. One of the neurological parameters analyzed in this study is serotonin. Stress is reported to be significantly mediated by serotonin (5-HT) and serotonergic pathways in the raphé nuclei of the brainstem where it innervates a wide range of structures involved in anxiety by acutely increasing 5-HT released at efferent targets (Hale and Lowry, 2011). Anxiety disorders may be caused by a variety of anomalies in the serotonergic system, including hypo- and hyper-innervation of important brain regions, and/or cellular processes resulting in aberrant neurotransmission. Hence, the aberrant regulation of 5-HT release and reuptake, or inappropriate response to 5-HT signaling, may be factors in the development of anxiety disorders. Low serotonin neurotransmission in depression supports the inflammatory hypothesis of depression. A decrease in serotonin production as noted by Coplan et al. (2014) was caused by the migration of tryptophan from serotonin biosynthesis to quinolinic acid production via the pro-inflammatory cytokines through the kynurenic acid pathway. Suffice it to say the drugs meant for treating mood disorders should exert bioactivity that modulates serotonergic pathways. In this investigation, PTZ induced a reduced serotonin level in the rats, and the studied extract mediated the alteration in the serotonin level. This serotonergic activity is relatively comparable to those induced by standard drugs like imipramine. Similarly, an increase in dopamine output was noticed in PTZ-induced animals' brains due to a reduction in GABAA receptor–mediated transmission. Exposure to various treatments results in stressful situations while the injection of anxiogenics like PTZ reduces GABAergic transmission and boosts dopamine production in rats moving freely, especially in the cerebral cortex and nucleus accumbens (Sharma and Parle, 2017). PTZ-induced rats demonstrated increased dopamine levels in this study, and this affirms a significant anxiogenic effect of the treatment as indicated by the increased locomotory activity of the rats in the OFT. Dopamine concentration significantly increased in the prefrontal cortex and nucleus accumbens (Nacc) of previously kindled rats, relative to the control group, after a subconvulsive dose (20 mg/kg) of PTZ was administered (Dazzi et al., 1997). Also, kindled rats' poor learning ability and the rise in locomotory activity following a challenge with subconvulsive PTZ doses were attributed to changes in dopaminergic neuronal regulation. The observed potential of the PN in counteracting the effect of a subconvulsive dose of PTZ may be due to its ability to restore biogenic amines, GABA, serotonin, dopamine, and noradrenaline levels in the brains of the experimental rats.

According to Charles et al. (1994), the brain of depressive individuals has higher levels of choline than the control animals while Higley and Picciotto (2014) observed that increased level of acetylcholine (ACh) can cause depressive symptoms in both people and animals. This trend was validated by Saricicek et al. (2012) who while using human imaging, noted that ACh levels are significant in individuals with a history of depression and even more elevated in actively depressed patients who showed signs of nicotinic receptors across the brain. Interestingly, this study supports the concept that linked depression and mania to changes in adrenergic-cholinergic equilibrium. Suffice it to say studies involving rat and human models also associated depressed symptoms with elevated cholinergic activity and reduced noradrenergic levels (Mineur et al., 2013). PTZ (20 mg/kg) was observed to elevate AChE activity in rats compared with the control corroborating Soares et al. (2018) who used flies. The differences in the PTZ active dose are a function of the body mass index of the animal model used and the drug pharmacokinetics (Choudhary et al., 2013). This is despite the spontaneous increase in the level of AChE activity as a physiological response to elevated ACh levels induced by the alteration of the cholinergic system in PTZ-treated experimental rats. Furthermore, AChE inhibitors often accelerate neurotransmission in the brain by slowing down the rate at which acetylcholine (ACh) is degraded. These observations support present findings that showed the inhibitory effect of the PN on acetylcholinesterase and suggest its possession of neuroprotective properties. Even though the mechanism underpinning the activity observed with the PN is not fully understood, it may be connected to its relative ability to inhibit the cholinergic system and prevent oxidative stress-related damage to the neurons.

5. Conclusion

The PN showed sedative, antidepressant, and anxiolytic properties in addition to improving the brain's antioxidant status via the stimulation of the neurotransmitters. Even though previous studies corroborated the results, the plant bioactivities were attributed to the interplay of phytochemicals, polyphenols, and other compounds. This supports the need for more investigations focusing on the use of PN fractions, their pure concentrates, toxicology and individual constituents to sufficiently validate the plant's neurotherapeutic properties. The complexity of the interaction between the phytoconstituents, the *in vivo* biochemical as well as pathophysiological responses, and how the interaction induces bioactivity are not yet fully understood. However, the study confirms that the PN bioactive phytochemicals interacted *in vivo* with the neurophysiological system to induce neuroprotective activities in the experimental rats. Although the potency of the plant's neurobiological activities may be less effective compared to standard clinical psychiatric drugs, it has insignificant side effects and promising prospects in psychiatric healthcare. The potent bioactivities such as cholinergic and monoaminergic modulatory activities, neurochemical modulatory activities, and brain-derived neurotrophic factor (BDNF) or neurotrophin regulatory processes that are observed during the study are further indications of the plant's suitability as a cost-effective, alternative resource for antidepressants, neurodegenerative and anxiolytic drugs. The effective application and ethnopharmacological relevance of *Parquetina nigrescens*, even in traditional healthcare systems, may be premised on the outcome of this study and future research that establishes pharmacological standards for its use in the treatment of neuropsychiatric conditions. This study validates the plant's ethnomedicinal status in treating illnesses related to stress, depression, and sleeplessness, and contributes to the existing data on the plant's therapeutic potential.

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CRediT authorship contribution statement

Ayokunmi Adebukola Akinduko: Conceptualization, Data curation, Funding acquisition, Investigation, Resources, Writing – review & editing. **Sule Ola Salawu:** Data curation, Funding acquisition, Project administration, Resources, Supervision. **Afolabi Clement Akinmoladun:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision. **Afolabi Akintunde Akindahunsi:** Conceptualization, Formal analysis, Investigation, Project administration, Supervision, Writing – original draft. **Osarenkhoe Omorefosa Osemwegie:** Conceptualization, Funding acquisition, Methodology, Resources, Writing – original draft, W.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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